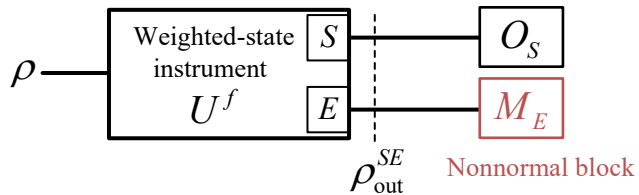
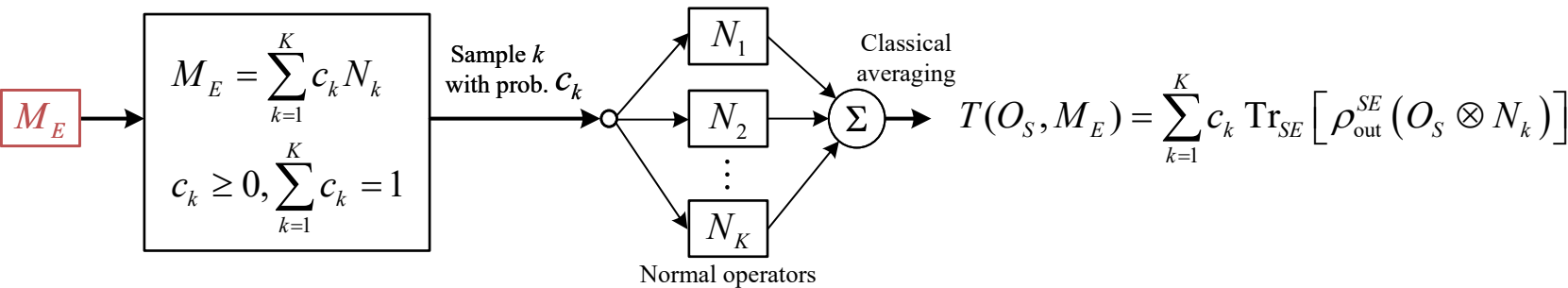


(a) Weighted-state framework and target

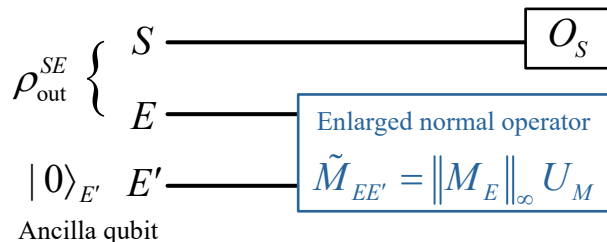


Target: $T(O_S, M_E) = \text{Tr}_{SE} \left[\rho_{\text{out}}^{SE} (O_S \otimes M_E) \right]$

(b) Existing realization: randomized normal decomposition



(c) Proposed realization: coherent Sz.-Nagy dilation



$T(O_S, M_E) = \text{Tr}_{SEE'} \left[\left(\rho_{\text{out}}^{SE} \otimes (|0\rangle\langle 0|)_{E'} \right) (O_S \otimes \tilde{M}_{EE'}) \right]$